Cancer Prediction using DNA Methylation Profiles

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Introduction

DNA methylation is one type of epigenetic marker. Epigenetic markers influence the activity of the genes without altering the DNA sequence and help the cells carry out their normal functions. DNA methylation is an important component in numerous cellular processes like chromosome separation during mitosis and meiosis. A reduction or increase in DNA methylation can be seen in cancerous cells.

The purpose of my project is to determine if the DNA methylation profiles of cancerous cells can help pinpoint a particular kind of cancer, from a set of possible cancers.

Background (Research)

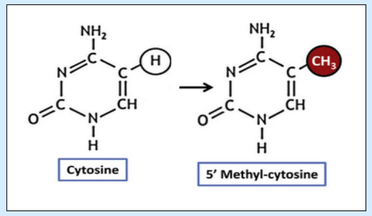
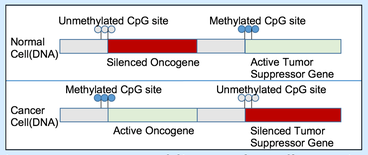


Image Credit: https://www.encyclopedie-environnement.org/

DNA methylation is the addition of a methyl(CH_3) group to the DNA strand. DNA Methylation pattern varies by cell types as gene expression varies by cell type.



Changes in regular DNA Methylation patterns like hypo-methylation or hyper-methylation can cause some genes to mutate to oncogenes and it can also silence tumor-suppressor genes. These changes can cause problems in regulating gene expression like growth signaling and apoptosis which in turn can cause cancer.

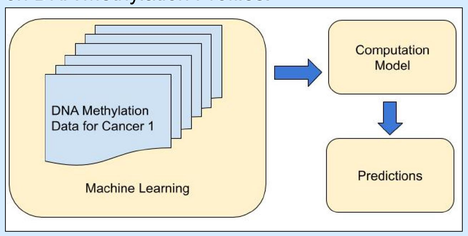
Research shows that DNA methylation patterns change **early** in some diseases such as cancer.

DNA methylation profiling can be done **non-invasively, quickly,** and **cost-effectively** compared to methods like biopsy.

Tools to help healthcare professionals identify a cancer type based on DNA methylation profiles are currently **unavailable**.

Research Goal

The goal of the research was to create a computational model using machine learning in order to predict a cancer type based on DNA Methylation Profiles.

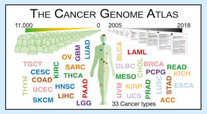


DNA Methylation profiles for the following cancers were used in this project: Acute Myeloid Leukemia, Sarcoma, Skin Cutaneous Melanoma, Stomach Adenocarcinoma, Lung Squamous Cell Carcinoma, and Brain Lower Grade Glioma.

The following multi-classification machine learning algorithms were used in this project: Decision Trees, Gaussian Naïve Bayes Classifier, Random Forest, Logistic Regression, and Support Vector Machine.

My project design criteria was that the precision, recall, and F1-Score of at least one of the machine learning algorithms in classifying different types of cancer should be 85% or better.

Data



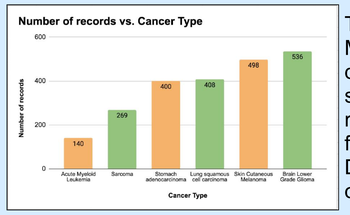
Freely available DNA methylation dataset from **The Cancer Genome Atlas** (TCGA) was downloaded and used in the project.

Image Credit: https://dinglab.wustl.edu/partnerships

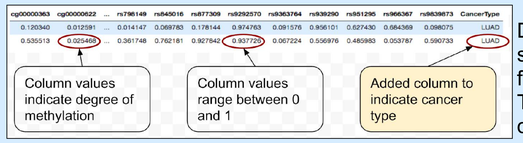
Data

No.	Cancer Type	Histological site
1	Acute Myeloid Leukemia	Blood forming tissue in bone marrow
2	Sarcoma	Connective tissues like bone, fat, muscle, cartilage
3	Stomach adenocarcinoma	Tissue cells that produce mucus
4	Lung squamous cell carcinoma	Tissue cells that form linings of internal organs
5	Skin Cutaneous Melanoma	Melanin forming cells
6	Brain Lower Grade Glioma	Tissues/Cells of the CNS

The project used 6 cancer types based on the 6 different cell tissue types the cancer originate's in.



The number of DNA Methylation records for different types of cancer are shown on the left. Each record had 485,000+ features which represents DNA methylation value at one of the CpG sites.



Data samples from the TCGA dataset.

Challenges:

- Data Imbalance: Records varied from 140 to 536 per cancer type.
- Dimensionality Problem: 2251 rows vs 485,000+ features.
- Missing or null values in records.

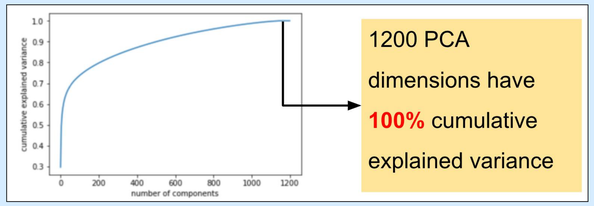
Procedure

Software:

Python, open-source packages such as scikit-learn and pandas, and Jupyter notebook

Solving Data Challenges:

- Used random sampling to select 200 records per cancer type.
- Used Principal Component Analysis (PCA) to reduce the dimension of the feature space to 1200 Principal Components.
- Used data imputation by mean values for missing data.

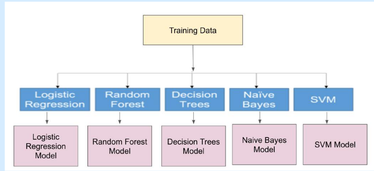


Data Wrangling results:

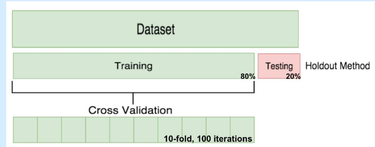
Cancer Type	# of records (pre processing)	# of records (post processing)	Sampling technique	# of columns (pre processing)	# of columns (post processing)
Acute Myeloid Leukemia (LAML)	140	200	Random over-sampling	485,578	1200
Sarcoma (SARC)	269	200	random under-sampling	485,578	1200
Stomach adenocarcinoma (STAD)	400	200	random under-sampling	485,578	1200
Lung squamous cell carcinoma (LUAD)	507	200	random under-sampling	485,578	1200
Skin Cutaneous Melanoma (SKCM)	475	200	random under-sampling	485,578	1200
Brain Lower Grade Glioma (LGG)	534	200	random under-sampling	485,578	1200

Each cancer type has 200 rows and 1200 features after data wrangling.

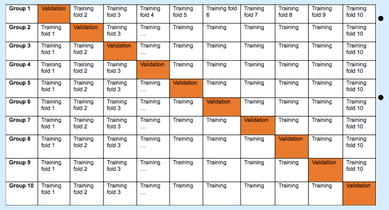
Procedure



During data download, the project labeled each record i.e. the cancer type for each record. Therefore used supervised machine learning algorithms.



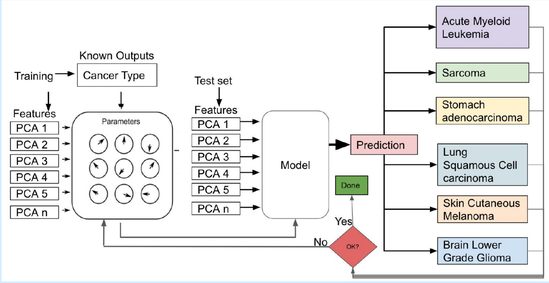
Data was split into a test set (20%) and train set (80%)



- Used 10 fold cross-validation for training to prevent overfitting.
- Final Validation accuracy = Average(Group1, Group2,..)

Pseudo-code:

- Model trained on training data by tuning the hyperparameters for algorithms.
- Tested the computational models with test data.
- Repeated building several models by tuning different hyperparameters until the results satisfied metrics criteria.



Metrics for Evaluating Models

Three **metrics**: Precision, Recall, F1_score

Four types of **errors**:

TN (True Negative): case was negative and predicted negative

TP (True Positive): case was positive and predicted positive

FN (False Negative): case was positive but predicted negative

FP (False Positive): case was negative but predicted positive

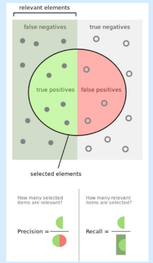


Image Credit: https://towardsdatascience.com/

Precision: Accuracy of positive predictions

Precision = TP/(TP + FP)

Recall: Fraction of positives that were correctly identified

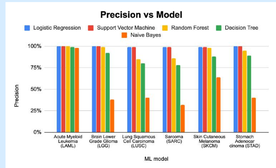
Recall = TP/(TP+FN)

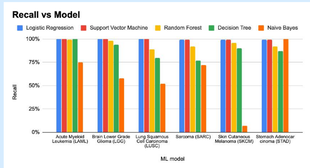
F1_score: Weighted harmonic mean of precision and recall

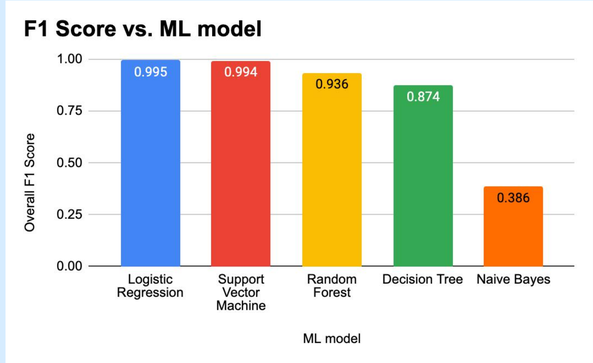
F1 Score = 2*(Recall * Precision) / (Recall + Precision)

Better predictor in analyzing the results as it seeks a balance between precision and recall

Results







Discussion

Model	Test-Statistic	Acute Myeloid Lymphoma (LAML)	Brain Lower Grade Glioma (LGG)	Lung Squamous Cell Carcinoma (LUSC)	Sarcoma (SARC)	Skin Cutaneous Melanoma (SKCM)	Stomach Adenocarcinoma (STAD)	Overall F1 Score
Logistic Regression	Precision	100%	100%	99%	99%	99%	100%	0.995
	Recall	100%	100%	100%	99%	99%	99%	
SVM	Precision	100%	100%	99%	99%	99%	100%	0.994
	Recall	100%	100%	100%	99%	99%	99%	
Random Forest	Precision	100%	99%	85%	86%	98%	95%	0.936
	Recall	99%	98%	89%	92%	96%	92%	
Decision Tree	Precision	99%	92%	80%	78%	88%	89%	0.874
	Recall	100%	94%	80%	77%	90%	87%	
Naive Bayes	Precision	98%	38%	40%	32%	64%	40%	0.386
	Recall	75%	58%	52%	72%	7%	1%	

- Gaussian Naïve Bayes generally performed poorly. Its mean precision and recall was 40% and it had an overall f1_score of 38.6%. Its poor prediction may be attributed to the algorithm's inherent assumption that all features are conditionally independent which might not have been true for this data.
- Decision Tree performed well with a mean precision and recall in the 85% and an overall f1_score of 87.4%.
- Random Forest performed better than Decision Tree with a mean precision and recall in the 93% and f1_score of 93.6%.
- Logistic Regression and Support Vector Machines performed the best producing precision, recall, and f1_score of over 99%.
- PCA turned out to be a good dimensionality reduction technique for DNA Methylation data.

Conclusions

- The results demonstrate that it is possible to **identify cancer types based on DNA Methylation profiles**.
- The best machine learning algorithms for this purpose were **Logistic Regression and SVM**.
- ML approach using DNA Methylation profiles can be used in **early cancer detection**.

Future Extensions

- Although the study utilizes only data for six cancer types, this study of machine learning-assisted decision-making can be extended to incorporate more cancers in the future.
- This dataset does not include DNA methylation data of normal cells. If the data becomes available in the future, this ML-based approach can be extended to see if it can predict normal vs cancerous cells based on DNA Methylation data.

Selected References

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3. Zhou, Weizhuang, et al. "Machine Learning Classification of Kidney and Lung Cancer Types ." CS229, Stanford University, 2013, cs229.stanford.edu/proj2013/JainMenZhou-MachineLearningClassificationofKidneyandLungCancerTypes.pdf. Accessed 20 Dec. 2018.